Alignment and Fusion Using Distinct Sensor Data for Multimodal Aerial Scene Classification

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***Abstract*— Multimodal sensors offer a wealth of rich and diverse data, which is helpful for classifying similar and complex aerial scenes. However, the heterogeneity of the data collected from different sensors brings a great challenge for alignment and fusion. For this, we present a multimodal aerial scene classification approach for extracting distinct modal information representations, realizing alignment and fusion of semantic information at both the data and feature levels. First, an adaptive zero-crossing rate (AZCR) module is proposed to convert the sequential data into images, achieving alignment at the data level. This module is proficient at extracting temporal and frequency domain features from sequential data through adaptive parameter adjustments. Second, we propose a multimodal alignment and fusion (MMAF) module to facilitate the alignment and fusion of distinct data, thereby achieving comprehensive modality integration at the feature level. Finally, the multimodal alignment loss (ALIGN Loss) function is designed to assess the alignment outcomes and constrain the training process. Our approach has been proven effective in accurately classifying aerial scenes, as demonstrated by the results of our experiments on two public datasets. The proposed method achieves 81.32% and 59.80% F1 score on the AuDio Visual Aerial sceNe reCognition datasEt (ADVANCE) and urban region function classification (URFC) datasets.**

***Index Terms*— Aerial scene classification, distinct sensor data, multimodal alignment, multimodal fusion.**

I. Introduction

AVIATION technology has made remarkable progress in recent years, leading to the wide use of aerial equipment like drones and aircraft. The task of aerial scene classification holds significant importance in various domains, such as land cover analysis [1], [2], urban planning [3], [4], and environmental monitoring [5], [6]. This task aims to assign scene labels to images by comprehensively assessing their content. While some scene classification methods [7], [8], [9], [10] use convolutional neural networks (CNNs) [11], [12], [13], [14] for extracting image features, they attempt to differentiate complex scenes based solely on visual information. As shown in Fig 1, there are four aerial image–audio pairs, and the audio data come from Freesound[1] and are preprocessed [22]. It is challenging to distinguish between forest and orchard [Fig. 1(a) and (c)], or train station and bridge [Fig. 1(b) and (d)] only based on visual cues. Fortunately, we find audio signals corresponding to these categories easier to distinguish. As a result, this article focuses on using different cues to identify complex scenes by leveraging complementary information such as audio signals (or other time-series data) and images from distinct sensors, aiming to improve aerial scene classification accuracy.

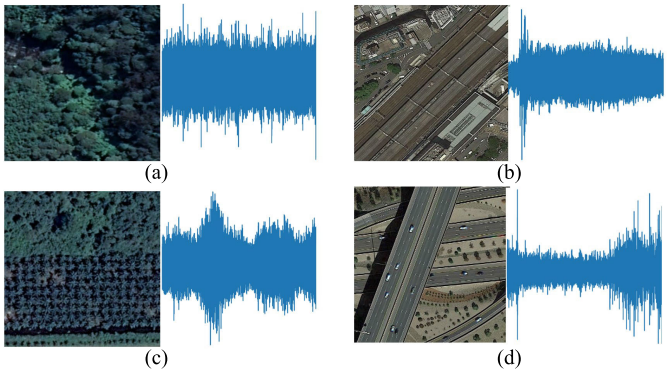


Fig. 1. Several pairs of audio-visual samples from the ADVANCE dataset. (a) and (c) are not easily distinguishable by separate visual features; the same applies to (b) and (d). Audio signals can help with scene classification.

Some existing methods [15], [16], [17] for multimodal data fusion adopt separate branch networks to extract features from heterogeneous modalities, followed by concatenation to obtain fused features. However, these approaches tend to overlook the inherent relationships that exist between different modalities. The misalignment between audio signals and images poses a significant challenge, which hinders effective feature interaction and fusion within the context of multimodal scene classification. A common approach is to convert audio into images to achieve data level alignment. This conversion usually involves the generation of audio spectrograms, which use methods such as log-mel spectrograms [16], [18] and mel-frequency cepstral coefficient (MFCC) [19]. These methods can be employed to generate spectral images encompassing frequency domain features. Zhang et al. [20] convert input audio into a log-mel spectrogram and treat it as a sequential segment-level sequence for scene classification using 3-D-CNNs. Similarly, in [21], the audio is converted into features through mel-scale filter banks, which transform each audio channel into a time-frequency domain representation.

However, the generation of spectrograms may result in the loss or blurring of specific features, potentially impacting the outcomes of audio signal analysis. To address this limitation, we apply the short-time zero-crossing rate, a common

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time-domain feature for audio signal processing, to analyze the rate of change and frequency characteristics of audio signals. We propose an adaptive zero-crossing rate (AZCR) module. This module is designed to extract the short-time zero-crossing rate by dynamically adjusting the window size. Combining the adaptive short-time zero-crossing rate with spectrograms to generate images can capture the crucial characteristics of time series data for aerial scene classification. The above process reflects the alignment of AZCR at the data level, which enhances flexibility in sensor selection and replacement for subsequent tasks. Data of distinct modalities can be transformed into a unified representation, effectively reducing the complexity of the subsequent processing. Subsequently, with the help of feature level alignment, merging features from different modalities can be more straightforward than data level alignment.

Consequently, after the initial data level alignment is achieved through AZCR, we propose a multimodal alignment and fusion (MMAF) module with separate backbones for each modality to extract features to achieve alignment at the feature level. The MMAF modules are embedded into different stages of the network for cross-modal information interaction, aligning and fusing information in the middle layer completely. Furthermore, we introduce a multimodal alignment loss (ALIGN Loss) function designed to evaluate the alignment between different modalities by narrowing down the gap between corresponding features to constrain the network training process and improve the robustness of the model.

This approach effectively integrates information from diverse modalities, enhancing aerial scene classification performance. We perform experiments on two multimodal datasets: AuDio Visual Aerial sceNe reCognition datasEt (ADVANCE) [22] and urban region function classification (URFC). Experimental results show the effectiveness of our multimodal aerial scene classification approach.

Our contributions can be summarized as follows.

1) Present an AZCR module to convert time-series data into images. This module can achieve initial alignment at the data level by adaptively adjusting the short-time zero-crossing rate (STZCR) parameters to suit different data.
2) Propose an MMAF module to align and fuse distinct modalities at the feature level by designing a novel ALIGN Loss to constrain the training process and evaluate the degree of alignment, minimizing the discrepancy between the feature representations of different modalities.
3) Experiments show that the proposed multimodal approach outperforms vision-only on two multimodal datasets. This approach consistently outperforms the existing state-of-the-art methods in terms of performance metrics.

II. Related Work

A. Time-Series Data Modeling

Time-series data modeling is a crucial task in data analysis, which involves the analysis and prediction of patterns within a sequence of data. This approach is especially relevant in audio signals, which are considered a type of time-series data. Audio signals frequently employ techniques like recurrent neural networks (RNNs) [23]. Although these techniques have driven significant advances, they face challenges related to vanishing and exploding gradients, which lead to the development of enhanced RNN models like long short-term memory (LSTM) [24] and gated recurrent unit (GRU) [25]. Transformer [26] enables learning dependencies across different positions within a series via a self-attention mechanism to capture relationships [27], [28], improving the ability to model the contextual information. However, it demands substantial computational resources and time, escalating training expenses. In speech recognition, numerous techniques are available for extracting temporal visual features from time-series data. These methods include gramian angular field (GAF) [29], MFCC [19], [30], and log-mel spectrograms [16], [31]. Nonetheless, generating GAF images can be computationally intensive, mainly when dealing with extensive time-series data. Spectrogram-based methods like MFCC and log-mel spectrograms are contingent upon time resolution, and the image quality is susceptible to variations based on the chosen parameters.

This article introduces the AZCR module, designed to dynamically adapt the parameters for extracting the STZCR. This adaptive methodology guarantees that the resulting images encompass pertinent time domain information, thereby augmenting the representation of temporal characteristics within the data.

B. Scene Classification

The discussion of scene classification methods is divided into two parts: RGB-based scene classification and multimodal scene classification.

1) RGB-Based Scene Classification: Over the past few years, significant advancements in scene classification have been achieved through CNNs [12], [32] and the availability of large-scale datasets [33], [34]. CNNs have proven effective in extracting multiple-level features from images [35], enhancing scene classification capabilities. Recent studies [36], [37] primarily rely on individual visual modalities using CNN-based methods for scene classification. In addition, with the vision transformer (ViT) [38] first applying transformer architecture to vision-related tasks, transformer-based [26] scene classification methods [39], [40] have emerged, exhibiting exceptional performance levels.

2) Multimodal Scene Classification: Given the challenges associated with distinguishing similar features in complex aerial and remote sensing scenarios relying only on image features, it is crucial to acknowledge the existence of various sensors [41] apart from cameras capturing RGB images. These sensors have the potential to provide additional data for enhanced scene analysis. As a result, multimodal scene classification surpasses RGB-based scene classification in such scenarios. In recent years, there has been a growing interest among researchers in scene analysis to explore the potential of multimodal fusion analysis. By leveraging the strengths of multiple modalities, this approach aims to achieve a more comprehensive and robust analysis of the scene under study.

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To a certain extent, the information acquired from depth cameras and acoustic sensors is valuable in distinguishing various scenes. Some methods [42], [43], [44] incorporate these modalities in conjunction with RGB images to enhance scene classification performance. Depth images are an instrumental source of information for determining the distance and depth of objects within a given scene. This attribute is highly beneficial in discerning distinct scene categories. DF^{2}Net [45] learns the discriminative feature representation and carries out feature fusion to realize the classification of RGB-D indoor scenes. Yuan et al. [43] propose an approach that utilizes adaptive cross-modal graph convolutional operations, capturing the correlation between the depth and RGB modalities efficiently.

Visual information has disadvantages, such as being easily sheltered, while sound information provides complementary insights [46], making audio-visual scene classification a promising approach for addressing these limitations. Hou et al. [44] design a dual-stream network that utilizes CNNs to process audio and visual inputs. This method can improve scene classification accuracy by leveraging the complementary information the two modalities provide. Zhou et al. [47] propose an attentional graph convolutional network (AGCN) to classify audio-visual scenes with structural perception. It can focus on semantically relevant and context-sensitive areas, beyond the previous CNN-based approach. Okazaki et al. [48] develop a CLIP late fusion network, representing the inaugural approach to employ CLIP [49] models for audio-visual scene classification tasks. Hu et al. [22] create the ADVANCE. This dataset includes mutual representations designed to uncover potential relationships between sound events and scenes and plays a pivotal role in pushing forward the field of aerial scene classification.

These methods are designed for specific modalities. In contrast, we develop a highly generalizable multimodal model that can be applied to a wide range of data types. Our approach aims to improve performance across different data types by integrating complementary information from various modalities, overcoming the limitations of traditional audio-visual scenarios.

C. Multimodal Alignment and Fusion

Alignment is a critical step in the process of the multimodal approach, which aims to ensure the coordination and consistency between different data sources to achieve the fusion of multimodal information effectively. Currently, two prevalent fusion approaches commonly apply in multimodal tasks: *early fusion* and *late fusion*. The dissimilarity of the two methods is determined by whether or not the fusion occurs before or after the alignment and interaction of the multimodal features. Early fusion methods [50], [51] consider integrating low-level semantic information. Late fusion methods [52], [53], [54] focus on feature processing after fusion by using concatenation or addition at the data level.

However, these methods need to pay more attention to the alignment relationship between the two modalities and sometimes introduce challenges like semantic conflicts and noise during the fusion process. These problems have been mitigated to some extent through attention mechanisms [55], [56] and contrastive learning methods [49], [57]. The classification framework proposed by Hong et al. [58] utilizes a deep encoder–decoder network to classify both HSI and LiDAR data. This framework emphasizes the reconstruction of input modalities from fused features to enhance the fusion level of extracted features. Liu et al. [59] propose a cross-modal fusion method called CMX. It corrects the features of two modalities through multilevel cross-channel interactions before performing the fusion operation. The approach proposed by López-Cifuentes et al. [60] relies on a semantics-driven attention mechanism to enhance the learning of relevant contextual information.

The modalities above correspond to the modalities of the visual sensor, which aligns spatial position information with RGB. In contrast, our approach does not restrict the modality format, allowing effective alignment and fusion even with distinct sensor data. Our approach follows a parallel architecture in the feature map processing branch, executing feature alignment and fusion at an intermediate layer. This strategy facilitates comprehensive and effective integration of various modalities. In addition, we introduce a multimodal ALIGN Loss that quantifies the degree of alignment and improves the training process without complicated model.

III. Proposed Method

As shown in Fig. 2, our approach comprises three essential components: an encoder for extracting AZCR, an MMAF module, and a multimodal ALIGN Loss function. To illustrate the approach, we specifically focus on two distinct modalities: time-series data and images. We take audio signals as an example of time-series data. The information on the proposed components and the overall optimization objectives are detailed in this section.

A. Encoder Architecture

When modeling time-series data like audio signals with RNN and LSTM, vanishing gradient issues may occur. These problems are often addressed by analyzing the generation of log-mel spectrograms. However, transformations like spectrograms may compress the data, potentially resulting in the loss or distortion of crucial features, which could impact analytical results. To overcome this limitation, we propose a feature compensation module incorporating learnable temporal domain information to improve the generated representation of sequence signals. Specifically, by introducing an adaptive parameter capable of dynamically adjusting the window size to extract the short-term zero-crossing rate feature, this can compensate for flaws resulting from spectrogram compression by automatically adjusting the generated features to the characteristics of the sequence signal. The proposed module is AZCR (see Fig. 2). For a detailed comprehension of the process as mentioned above, the corresponding expression is as follows.

Let $x(t)$ represent an audio signal, where t denotes time. In audio signal processing, $x(t)$ is segmented into frames. For any $l(l \in [1, L])$ sampling points within an L frame duration,

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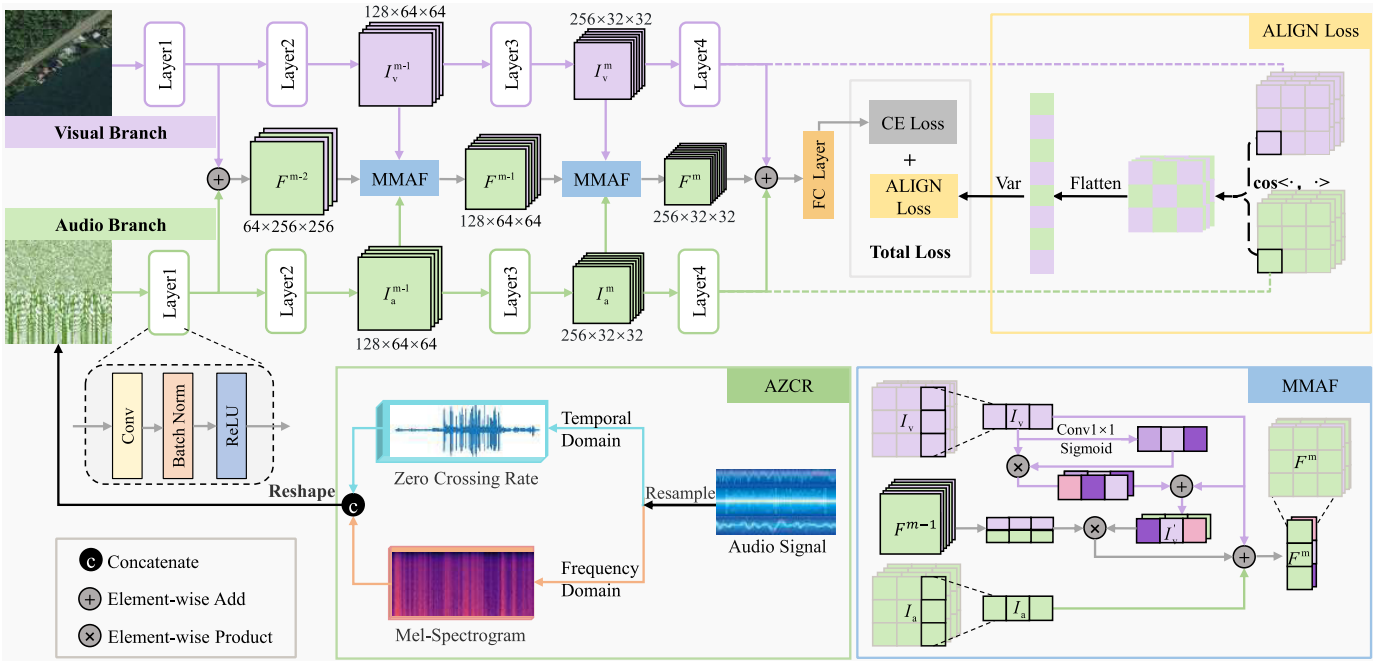


Fig. 2. Overview of our distinct sensor data multimodal classification method, which consists of an encoder with extracting AZCR, an MMAF module, and a multimodal ALIGN Loss.

the STZCR of this frame can be calculated as the following equation:

$$\mathrm{STZCR} = \frac{1}{L}\sum_{l=1}^{L}\mathbb{I}(x(l)x(l-1) < 0) \tag{1}$$

where L denotes the window size, determining the number of sampling points within each frame. $\mathbb{I}$ is the indicator function, and we define a set of learnable parameters $\boldsymbol{w} = [w_1, w_2, \ldots, w_n]^T$, where $n \in Z+$ represents the set of positive integers. The dimension n of this vector can be adjusted based on the specific context. Subsequently, we utilize the parameter vector $\boldsymbol{w}$ to update L to calculate the STZCR. During each training iteration, $\boldsymbol{w}$ is updated as follows:

$$\boldsymbol{w}' = \boldsymbol{w} - \eta\frac{\partial \boldsymbol{w}}{\partial w_{\text{time}}} \tag{2}$$

$$w_{\text{time}} = \sum_{i=1}^{n} w_i' \tag{3}$$

where η is the learning rate; after the parameters are automatically updated, (1) can be written in the following form to implement the calculation of AZCR:

$$\mathrm{AZCR}(x(t)) = \frac{1}{w_{\text{time}} * L}\sum_{l=1}^{L}\mathbb{I}(x(l)x(l-1) < 0) \tag{4}$$

$$I_a = [\mathrm{Mel}(x(t)); \mathrm{AZCR}(x(t))]. \tag{5}$$

The symbols $\mathrm{Mel}(\cdot)$ and $\mathrm{AZCR}(\cdot)$ signify the techniques for extracting frequency and temporal domain features, respectively. Following the completion of processing in both frequency and temporal domain branches, we employ a concatenation operation denoted as [;] to merge the branches' outputs, thereby generating the feature map.

The process of generating a feature map from a sequence signal can be summarized as (5). The data representation generated by this module combines features from both the frequency and temporal domain in the time-series data, obtaining $I_a \in \mathbb{R}^{H\times W\times C}$ of the feature map with height H and width W, and C is the number of input channels into the convolutional neural network. Subsequently, the resulting feature map I_a is fed into one branch of multimodal processing.

B. Multimodal Alignment and Fusion

The transformation of audio signals into images at the data level alignment fails to ensure complete alignment between them. Therefore, following the generation of image I_a through AZCR, it is necessary to align and merge it with another modality at the feature level. As illustrated in Fig. 2, we design the MMAF module to address the challenges of alignment and fusion posed by distinct information.

This module facilitates effective interaction between different modalities, and integrating information from various data sources enhances the effectiveness of data fusion. Inspired by [61], the CNN-based feature extractor for each modality independently is designed. Given that RGB images contain a more extensive range of color and texture information compared with other modalities, the visual branch is designed with a more complex structure than another branch. The process can be outlined as follows:

$$I_v' = I_v \otimes \sigma[\phi(I_v)] \oplus I_v \tag{6}$$

where $I_v \in \mathbb{R}^{H\times W\times C}$ is the feature map from the visual branch. The symbol $\otimes$ represents elementwise matrix multiplication, $\oplus$ represents elementwise matrix addition, σ represents the sigmoid function, and ϕ represents the 1×1 convolution operation. Combining 1×1 convolution with a sigmoid activation function in this module can enrich feature information within the visual branch, promoting channel mixing and improving image representations. Using a 1×1 convolution transforms

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the feature map's channel dimension while applying a nonlinear activation function enhances its capacity for nonlinear representation, allowing more effective modeling of intricate data distributions. Subsequently, the visual branch representation ($I'_v \in \mathbb{R}^{H\times W\times C}$) and the audio branch representation ($I_a \in \mathbb{R}^{H\times W\times C}$) are fused. This process can be expressed as follows:

$$F^m = F^{m-1} \otimes I'_v \oplus I_a \oplus I_v \tag{7}$$

where $F^m \in \mathbb{R}^{H\times W\times C}$ is the mth fusion feature map ($m = 1, 2, \ldots, N$) and N denotes the number of layers in ResNet [12]. The MMAF module interacts among features from modalities and within a single modality. As shown in Fig. 2, the $\otimes$ operation aims to highlight and align relevant information, and the $\oplus$ operation leads to the fused feature. The MMAF module stands out in its ability to integrate and facilitate interactions between information originating from various modalities, all while preserving the unique semantics within each modality.

C. Loss Function

During the model optimization process, the total loss function comprises two components: the classification loss and the multimodal ALIGN Loss.

1) Classification Loss: With the goal of ariel scene classification, the cross-entropy loss (L_{CE}) is used to measure the discrepancy between predicted results and ground truth labels. The cross-entropy loss of the network in this article is

$$L_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C}\left(t_{i,j}\cdot\log\left(\frac{\exp(y_{i,j})}{\sum_{k=1}^{C}\exp(y_{i,k})}\right)\right) \tag{8}$$

where C and N denote the number of classes and samples in the batch, respectively. $y_{i,j}$ denotes the score for class j for the ith sample, t represents the true class labels for the samples in one-hot encoded form, and $t_{i,j}$ equals 1 if the ith sample belongs to class j and 0 otherwise.

2) Multimodal ALIGN Loss: To evaluate the alignment between the modalities, we design a multimodal ALIGN Loss (L_{ALIGN}), which can be expressed as

$$L_{\text{ALIGN}} = \text{var}\left(J - \frac{Z_v Z_a}{\|Z_v\|\cdot\|Z_a\|}\right) \tag{9}$$

where Z_v and Z_a denote a pair of feature tensors from the final convolutional layer. We refer to their l_2-norm as $\|\cdot\|$. J denotes an all-1 matrix. The multiplication of similarity between two feature maps can be used to assess their alignment. Suppose the result of the elementwise multiplication between the corresponding elements in two feature maps approaches 1. In that case, it indicates a high similarity between the feature maps or alignment of the features in their respective positions. Subsequently, the resulting matrix is subjected to the var$(\cdot)$ operation, which calculates the variance of all the elements in the resulting matrix, yielding a numerical value that serves as the ALIGN Loss function's value.

After the fusion of features from different modalities by the MMAF module, these features are fed into the classification head composed of fully connected layers. Therefore, the classification and multimodal alignment losses are used jointly to optimize the model. The overall objective function is formulated as follows:

$$L_{\text{total}} = L_{CE} + \lambda L_{\text{ALIGN}} \tag{10}$$

where λ is a balance weight and its specific impact on the model's performance will be thoroughly discussed in the experimental section of our study. By designing this multimodal ALIGN Loss, we ensure that both branches learn complementary features from different modalities during the joint training process. This approach effectively mitigates the issue of a single modality being overlooked or challenging to optimize, ultimately enhancing the fusion of modalities and overall model performance.

TABLE I
DESCRIPTION OF THE ADVANCE DATASET, INCLUDING THE NAME OF THE CLASS AND THE CORRESPONDING COUNT OF SAMPLES FOR TRAINING, VALIDATION, AND TESTING SETS

No.	Class Name	Train	Validation	Test
0	grassland	105	15	30
1	sparse shrubland	280	40	85
2	bridge	195	25	60
3	harbour	353	50	105
4	orchard	139	17	51
5	beach	151	21	44
6	airport	125	15	45
7	lake	250	35	72
8	residential	739	105	212
9	train station	180	25	55
10	farmland	300	40	90
11	sports land	109	8	42
12	forest	603	86	173
	Total	3529	482	1064

IV. EXPERIMENTS

A. Dataset

To assess the effectiveness of the suggested approach for the classification of aerial scenes, we utilize two multimodal datasets: ADVANCE [22] and URFC.

The ADVANCE dataset in summary contains 5075 pairs of audio-visual, involving 13 aerial scene classes, i.e., grassland (gra.), sparse shrubland (spa.), bridge (bri.), harbor (har.), orchard (orc.), beach (bea.), airport (air.), lake (lak.), residential (res.), train station (tra.), farmland (far.), sports land (spo.), and forest (for.). Each audio-visual pair consists of an RGB image of 512 × 512 pixels resolution and a 10-s audio clip. The audio data are collected from Freesound, and then recordings are from 2-10 s to 10 s by copying the audio content. In our experiment, the allocation of audio-image pairs for training, validation, and testing is based on the split ratio outlined in [22]. For testing, 20% of the pairs are used, while 70% are for training and 10% for validation. The specific dataset partitioning is randomized for each experiment. Table I describes each scenario category and the number of samples.

The URFC dataset from the International Big Data Contest 2019 consists of some pairs of time-series data of an hourly aggregated number of user visits and remote sensing images, with labels of nine categories, i.e., residential area (res.),

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TABLE II
DESCRIPTION OF THE URFC DATASET, INCLUDING THE NAME OF THE CLASS AND THE CORRESPONDING COUNT OF SAMPLES FOR TRAINING, VALIDATION, AND TESTING SETS

No.	Class Name	Train	Validation	Test
0	residential area	7547	937	937
1	school	5923	758	758
2	industrial park	2847	384	384
3	railway station	1073	139	139
4	airport	2818	327	327
5	park	4433	526	526
6	shopping area	2795	347	347
7	administrative district	2075	253	253
8	hospital	2273	302	302
	Total	31784	3973	3973

school (sch.), industrial park (ind.), railway station (rai.), airport (air.), park (par.), and shopping area (sho.). There are two subsets of URFC datasets, URFC-A and URFC-B [15]. For our research, we utilize 39 730 samples from the URFC-A training set and divide the dataset into training, validation, and testing sets with an 8:1:1 ratio. Table II describes each scenario category and the corresponding number of samples available in this dataset.

B. Experimental Setup

1) Implementation Details: The experiments are implemented with the Pytorch [62] on a single NVIDIA GeForce RTX 3090 GPU. The backbone is ResNet, which includes four levels of blocks with channels 64, 128, 256, and 512. We employ the pretrained ResNet18 and ResNet50 [12] for training. The training batch size is 16, and the initial learning rate is 1e-5, decreasing by 0.8 after every five epochs for the ADVANCE dataset. The Adam optimizer [63] is used for backpropagation in the experiments.

2) Evaluation Metric: We adopt four widely recognized metrics to evaluate the classification results: accuracy, precision, recall, and F1 score. In specific terms, accuracy is defined as the ratio of correctly classified samples to the total number of samples within a dataset

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{11}$$

where TP and TN are the number of correctly predicted positive and negative instances, correspondingly. FP is the number of incorrectly predicted positive instances; TN is the number of correctly predicted negative instances. n represents the number of categories in the classification task. Accuracy is only a portion of the performance evaluation for a classification model. It should also be combined with other metrics like precision, recall, and the F1 score. The specific calculations for these essential metrics are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{12}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

$$\text{F1 score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{14}$$

Precision is the ratio of truly positive samples among the samples predicted as positive. Recall is the ratio of truly positive samples to the total number of positive samples on the dataset. F1 score is a metric that combines precision and recall into a single measure. It is defined as the harmonic mean of precision and recall, which provides a balanced evaluation of precision and recall and allows for expressing different preferences for these two metrics. Due to the differences between the number of categories, in this setting, the F1 weighted metric is used. The resulting accuracy and recall values are almost identical. Precision is the ratio of truly positive samples among the samples predicted as positive. Recall is the ratio of truly positive samples to the total number of positive samples on the dataset. F1 score is a metric that combines precision and recall into a single measure. It is defined as the harmonic mean of precision and recall, which provides a balanced evaluation of precision and recall and allows for expressing different preferences for these two metrics. Due to the differences between the number of categories, the F1 weighted metric is used. The resulting accuracy and recall values are almost identical, in this setting.

C. Results and Analysis

1) Comparison With State-of-the-Art Methods: To demonstrate the efficiency of the suggested approach, we select several state-of-the-art multimodal scene classification algorithms for comparison, which include multimodal fusion transformer (MFT) [42], semantic-aware scene recognition [60], and cross-task transfer [22]. Table III compares the accuracy and F1 score of the ADVANCE and URFC datasets with other scene classification methods. Classes 0–12 correspond to the Class Name columns of Tables I and II.

As shown in Table III, for the ADVANCE dataset, our method performs well in most classes, especially in Class 6, achieving 100% accuracy. MFT [42] and semantic-aware [60] methods performed relatively poorly in different categories, but the latter method has higher accuracy in some categories, such as Class 8 and Class 12. Cross-task transfer [22] also performs relatively well in most categories, but there are still some gaps relative to our method. Only one of the 13 categories outperforms our approach. For the URFC dataset, our method performs well in several classes, and results in Class 2 and Class 6 surpass those of other methods. MFT [42] and semantic-aware [60] work well in some categories, but less well in others. The performance of the cross-task transfer [22] is more consistent across all categories and slightly lower than that of our method in general. In summary, our method achieves better overall accuracy and F1 weighted score results than other methods.

After the quantitative analysis of the above indicators, the class activation map (CAM) [64] method is used in Fig. 3 for qualitative analysis. The aim of utilizing CAM in scene classification is to aid in comprehending the decision-making process and areas of emphasis in the classification process of a network through the display of the distribution of a particular category's feature on an image. In Fig. 3, the first row represents the "bridge" category, while the second row

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TABLE III
COMPARISON OF OUR APPROACH WITH OTHER METHODS ON THE ADVANCE AND URFC DATASETS FOR SCENE CLASSIFICATION. "ACC" REFERS TO ACCURACY AND "F1-W" REFERS TO F1 WEIGHTED SCORE

		ADVANCE				URFC			
Metric	Category	MFT [42]	Semantic-Aware [61]	Cross-Task Transfer [22]	Ours	MFT [42]	Semantic-Aware [61]	Cross-Task Transfer [22]	Ours
Acc(%)	Class0	0.00	0.00	13.33	**20.00**	**77.96**	29.93	50.57	74.11
	Class1	31.37	54.12	51.76	**88.24**	44.47	**59.15**	54.11	56.18
	Class2	19.44	10.00	43.33	**81.67**	21.67	25.62	26.17	**44.78**
	Class3	40.32	50.48	61.90	**68.57**	0.00	7.26	**37.90**	35.77
	Class4	3.27	0.00	39.22	**47.06**	62.54	70.90	63.47	**73.87**
	Class5	46.21	4.55	68.18	**75.00**	**57.82**	44.34	60.67	54.62
	Class6	18.52	13.33	75.56	**100.00**	11.30	15.07	42.61	**50.00**
	Class7	71.30	75.00	65.28	**90.28**	10.75	0.00	**40.86**	34.08
	Class8	88.36	**99.06**	97.17	98.11	24.39	8.01	**62.37**	46.04
	Class9	0.00	9.09	54.55	**72.73**	-	-	-	-
	Class10	47.04	57.78	52.22	**72.22**	-	-	-	-
	Class11	0.00	0.00	**42.86**	14.29	-	-	-	-
	Class12	78.03	**93.64**	87.86	93.06	-	-	-	-
	Overall	49.81	56.02	68.82	**79.70**	44.65	37.27	53.21	**59.24**
F1-w(%)	**Overall**	44.14	64.48	67.79	**81.32**	39.87	34.68	53.57	**59.80**

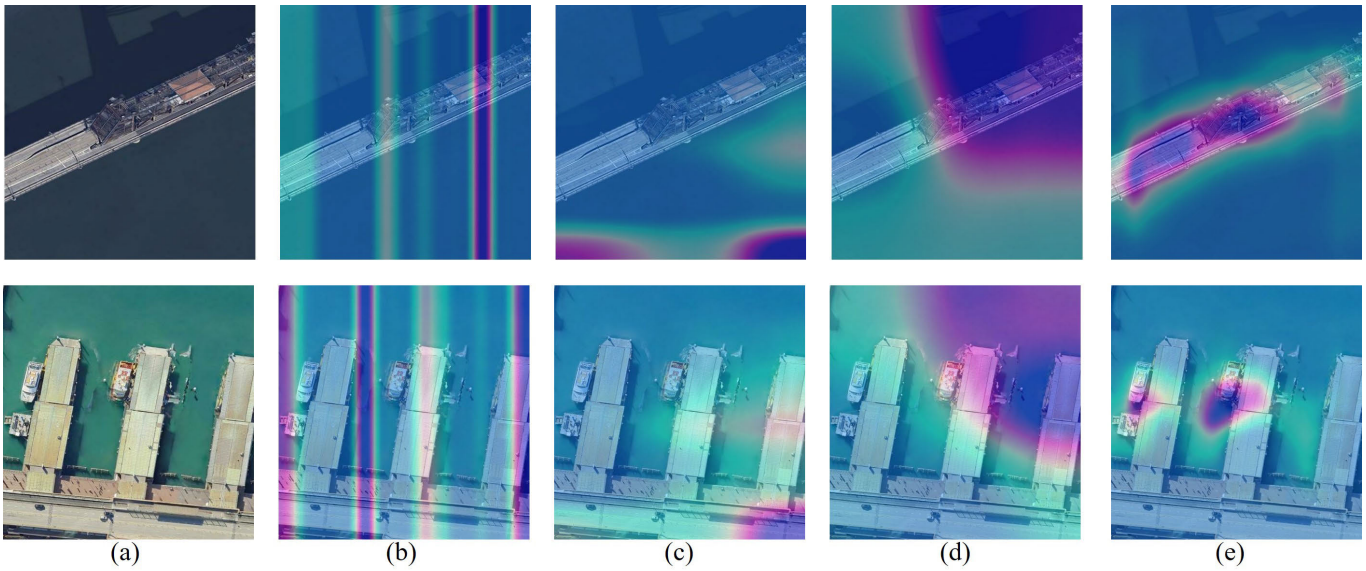


Fig. 3. Visualization results of the CAM on the ADVANCE dataset. (a) Input RGB images. (b)-(d) Results of MFT, cross-task transfer, and semantic-aware scene recognition. (e) Results of our method on scene classification.

represents the "harbor" category. From the results of CAM, it is evident that our approach [Fig. 3(e)] can highlight more distinctive features, leading to a higher predicted probability for the correct classes than other methods. It is worth noting that our method can improve classification performance to some extent but may bring more expensive computational costs. This is because when the multimodal framework handles distinct data, the parameters of each branch must be learned according to the characteristics of the modality and cannot be the same, which leads to a large number of parameters.

2) Comparison With Different Fusion Methods: We evaluate our proposed fusion method alongside early and late fusion methods. The MMAF module is an improved version of the middle fusion. Table IV lists the results of various fusion techniques concerning three measures: precision, recall, and F1 score. Our findings reveal that our method consistently outperforms early and late fusion methods across all evaluation metrics. This superiority can be attributed to our approach, which facilitates deep level feature interactions through a multiscale intermediate feature interaction approach, thereby improving the performance of the scene classification task.

Table IV illustrates that simple fusion methods (early and late fusion) may yield little benefits for aerial scene classification tasks. The inherent disparities between the original features of different data modalities make early fusion impractical. Late fusion projects feature in a higher-dimensional space without appropriate alignment, leading to insufficient information fusion. In contrast with other methods, our approach employs an intermediate layer feature alignment and fusion module to enhance classification accuracy by maximizing the use of intermodal features for higher classification performance.

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TABLE IV

COMPARISON ANALYSIS OF MMAF WITH EARLY AND LATE FUSION METHODS ON ADVANCE DATASET. "PRE" REFERS TO PRECISION, AND "REC" REFERS TO RECALL. "F1-W" REFERS TO F1 WEIGHTED SCORE. ACCURACY AND RECALL VALUES ARE EQUAL IN THE SETTING OF WEIGHTED FOR F1

Model	Fusion Method	Pre(%)	Rec(%)	F1-w(%)
ResNet18	Early (concat)	68.47	61.00	63.57
	Late (concat)	68.54	57.61	60.62
	MMAF	**84.22**	**78.95**	**80.55**
ResNet50	Early (concat)	68.17	61.18	63.49
	Late (concat)	62.19	51.79	55.44
	MMAF	**78.73**	**75.38**	**75.25**

TABLE V

ABLATION STUDIES OF DIFFERENT MODULES ON THE ADVANCE AND URFC DATASETS. "PRE" REFERS TO PRECISION, AND "REC" REFERS TO RECALL. "F1-W" REFERS TO F1 WEIGHTED SCORE. ACCURACY AND RECALL VALUES ARE EQUAL IN THE SETTING OF WEIGHTED FOR F1

Dataset	Method	Acc(%)	Pre(%)	F1-w(%)
ADVANCE	Baseline	77.63	83.85	79.59
	Baseline+AZCR	78.57	84.28	80.25
	Baseline+MMAF	78.63	**84.57**	79.88
	Baseline+AZCR+MMAF	**78.94**	84.22	**80.55**
URFC	Baseline	55.75	56.76	55.94
	Baseline+AZCR	58.19	59.39	58.52
	Baseline+MMAF	54.97	**60.46**	55.62
	Baseline+AZCR+MMAF	**58.38**	59.06	**58.89**

D. Ablation Study

1) Evaluation of the Proposed Modules in Our Method: We perform ablation experiments on the ADVANCE and URFC datasets to validate the AZCR and MMAF modules for multimodal aerial scene classification. The experimental results are shown in Table V. The baseline method employs two branches, both utilizing the ResNet18 architecture, and the fusion strategy employed is the traditional summation method, where the outputs of two branches are added together.

Using the AZCR module leads to varying degrees of enhancement in accuracy, precision, and F1 score. This improvement can be attributed to the enhanced feature extraction of temporal signals through the adaptive extraction of short-term zero-crossing rates. With the MMAF, apart from slight improvements in accuracy and F1 score, the best result was achieved with a precision of 84.57%. The combined effect of the two modules can lead to the highest accuracy and F1 score for classification. Regarding the limitations of AZCR, it could be argued that combining two modules leads to lower precision. One possible reason is that the ratio of the various categories is unbalanced, and the improvement of the model may be affected. In this case, if the model is more inclined to predict a certain category, even if the accuracy increases, it may cause the precision to decrease.

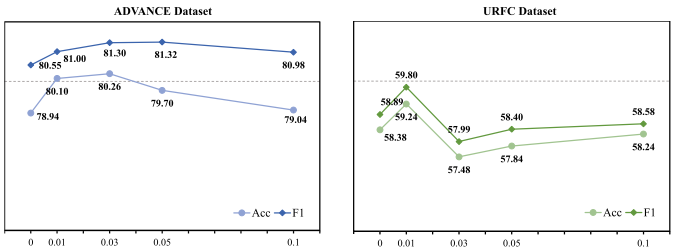


Fig. 4. Assessment of the impact of weight loss parameter λ on ADVANCE and URFC datasets. The axis scale represents the value of λ.

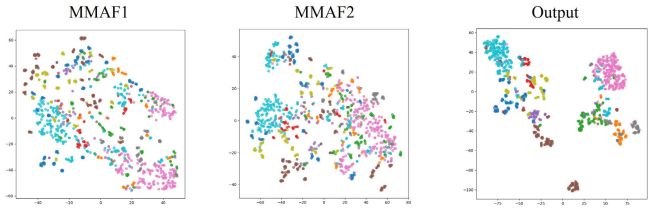


Fig. 5. Visualizing t-SNE representations of extracted deep-level features. The points in different colors represent distinct scene categories.

2) Evaluation of λ: The parameter λ serves as a balance weight of cross-entropy loss and our multimodal ALIGN Loss. The influence of different values of the loss weight on the experimental outcomes is depicted in Fig. 4. We analyze the assessment ranges from 0 to 0.1 here. As shown in Fig. 4, $\lambda = 0.03$ obtains the highest accuracy for the ADVANCE dataset, and $\lambda = 0.05$ achieves the best F1 score. For the URFC dataset, $\lambda = 0.01$ obtains the best accuracy and F1 score in the aerial scene classification task. Through comprehensive evaluation of both F1 score and accuracy, we find that it is comprehensively considered to set λ between 0.01 and 0.05, yielding a relatively stable performance across the range.

E. Visual Analysis

In this section, we employ two visualization methods, t-stochastic neighbor embedding (t-SNE) [65] and the confusion matrix, to qualitatively assess the proposed method. The confusion matrix offers a holistic view of the classification performance, and t-SNE visualizes the high-dimensional feature embeddings in a lower-dimensional space for better understanding and analysis.

1) Visualization of t-SNE: The t-SNE is used to visualize the feature representations obtained through network layers as a 2-D distribution. As described in Section III-B, the fusion at each intermediate layer can be achieved using the MMAF module. Fig. 5 demonstrates the visualization results of features at the different convolutional layer positions. The results corresponding to MMAF1 and MMAF2 represent the feature distribution of each category after different numbers of multimodal fusion and alignment modules. The visualization of output layer feature distribution illustrates that MMAF facilitates the extraction of deep-level features, which helps increase the interclass distance and decrease the intraclass distance.

2) Visualization of Confusion Matrix: The confusion matrix illustrates the association between the model's predictions and the true labels. Darker colors along the diagonal represent

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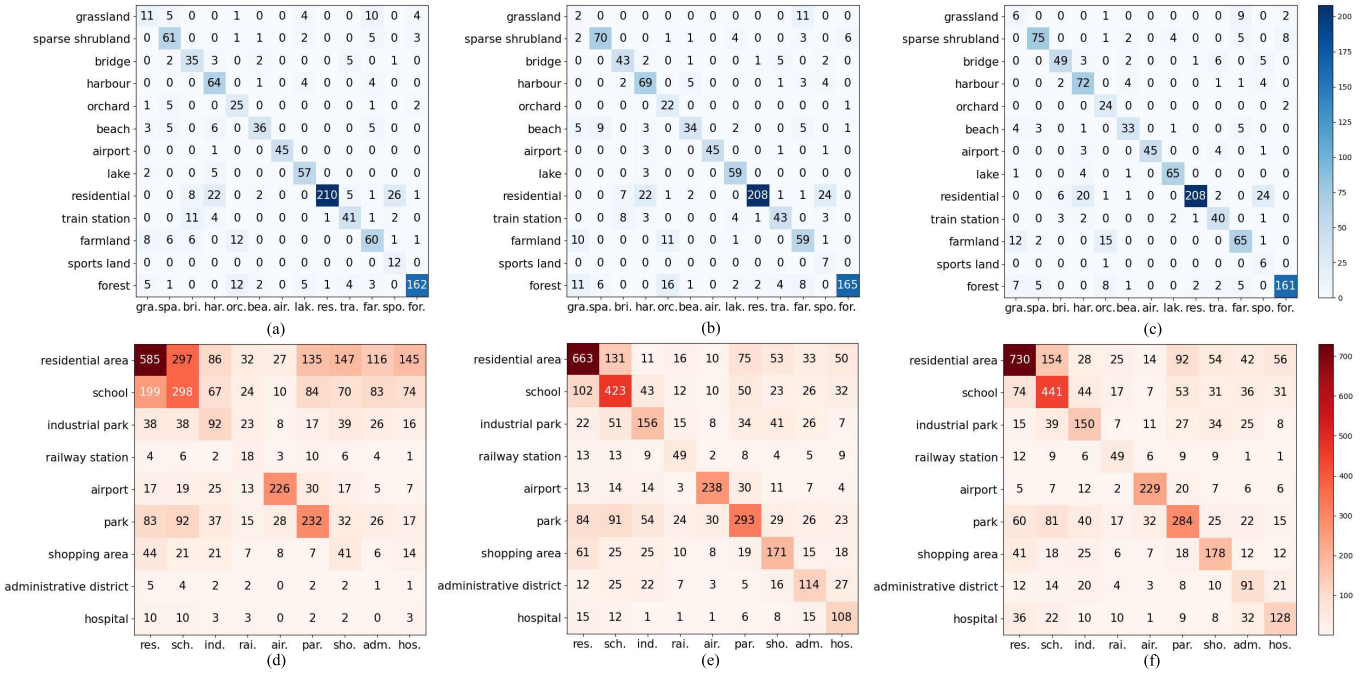


Fig. 6. Confusion matrix on the ADVANCE (the first row) and URFC (the second row) datasets. (a) and (d) Unimodal method with only images. (b) and (e) Baselines for the multimodal method. (c) and (f) Method proposed in this article. Due to space constraints, labels on the X-axis refer to the first three letters of the class name as abbreviations.

more accurate classification outcomes. Due to space constraints, labels on the position of true labels refer to the first three letters of the class name as abbreviations. Fig. 6 compares the confusion matrix results for the unimodal, multimodal baseline, and proposed method. It is evident that specific categories, like "school," pose challenges regarding potential confusion when relying only on image information. Compared with the unimodal approach that only considers visual features, the multimodal approach, which incorporates features from other modalities, enables better discrimination of specific categories, especially the "sparse shrubland" and "bridge" on the ADVANCE dataset, and the "administrative district" and "hospital" on the URFC dataset. Our approach achieves better performance by effectively facilitating interaction between features, resulting in optimal performance.

The limitations of our approach can be seen from the confusion matrix. Classes like "airport," "sports land," and "grassland" have limited samples in the ADVANCE dataset, leading to poor classification results. Similar challenges arise for the "railway station" and "administrative district" classes on the URFC dataset. A class with fewer data may have fewer samples to cover its entire distribution in feature space, and the model may be unable to learn enough feature representations from a limited number of samples.

V. Conclusion

This article introduces an approach for aligning and fusing multimodal distinct sensor data to improve aerial scene classification. The proposed method is composed of three components: a sequence signal modeling module named AZCR, a multimodal alignment and fusion module named MMAF, and a multimodal ALIGN Loss function. AZCR can convert the sequential data into images by extracting adaptive features to achieve alignment at the data level. The MMAF module is used to align and interact with features from distinct data in the middle layers of the network, enabling comprehensive modality fusion at the feature level. The training process is optimized using a multimodal ALIGN Loss function, which improves alignment effectively to blend into representations more conducive to scene classification. To evaluate the effectiveness of our approach, we conducted experiments on two public datasets. These results demonstrate the potential of our approach in enhancing the classification accuracy and overall performance in aerial scene classification.

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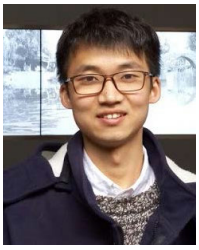
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